



CHEMISTRY CH.7 — PERIODIC TABLE — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

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1. Periodic Table: Development & Classification History

- Elements ko classify karne ke pehle attempts:
 - Dobereiner's Triads: 3 elements jinki atomic masses nearly same proportion mein hain; middle element ka mass average hota hai. Example: Li (7), Na (23), K (39). Cl-Br-I triad: $(35.5+126.92)/2 \approx 79.9$ (actual), calculated ≈ 81.2 . Known triads: Li-Na-K, Ca-Sr-Ba, Cl-Br-I — N-P-Sb NAHI hai (middle should be As, P fit nahi karta). Defect: sirf 3 triads identify kar paaye, baaki known elements fit nahi hue. [Q648, Q650, Q651, Q657, Q675, Q689, Q723, Q735, Q962]
 - Newlands' Law of Octaves: Har 8ve element ka properties pehle se repeat hoti hai; music ke octaves jaisa. Limitation: Sirf calcium tak kaam karta hai; heavier elements fit nahi hote; Cobalt aur Nickel same slot mein rakhe. Be aur Mg same properties dikhte hain (Mg, Be ke baad 8th element). 10th element 3rd element (10-7) jaisa resemble karta hai. Noble gases us waqt discovered nahi the isliye include nahi hue. Last element jo Newlands ne classify kiya woh Thorium tha (56 elements list mein). [Q650, Q652, Q736, Q737, Q738, Q743, Q806, Q808, Q978, Q990]
 - Mendeleev's Periodic Table: Atomic mass ke basis par arrange kiya; gaps chhodi future elements ke liye (gallium=Eka-aluminium, germanium=Eka-silicon, scandium=Eka-boron). Merits: Predictive power (oxides bhi predict kiye — Eka-aluminium ka oxide Ga_2O_3); demerits: isotopes ko separate position nahi mili (yahi uski asli limitation thi, "correct isotope position" uska feature NAHI tha), hydrogen ki fixed position nahi thi, some elements order galat (Ar before K, Co before Ni rakha gaya despite atomic mass higher hone ke). Mendeleev ne 63 known elements se shuru kiya tha; 8 groups aur 7 periods banaye the; H aur O ke saath compounds par focus kiya kyunki wo bahut reactive hain. Baad mein discover hue elements: Sc, Ga, Ge (predicted gaps fill kiye). [Q652, Q654, Q656, Q663, Q669, Q713, Q744, Q746, Q776, Q782, Q800, Q812, Q858, Q870, Q874, Q898, Q954, Q964,

- Modern Periodic Law (Moseley, 1913): Elements ki properties unke atomic number ka periodic function hain; ismein elements ki increasing atomic number (nuclear charge) ke order mein arrange kiya jaata hai — Mendeleev ke atomic mass wale order se ye alag hai. Neutral atom mein atomic number = protons ki sankhya = electrons ki sankhya (koi net charge nahi hota). In early classification attempts, elements ko unki properties ke basis par classify karne ki koshish hoti thi. [Q654, Q656, Q709, Q862, Q882, Q727]
- Modern Periodic Table mein 7 periods (horizontal rows) aur 18 groups (vertical columns) hain. [Q656, Q658]

2. Modern Periodic Table Structure

- Period = horizontal row; left to right atomic number badhta hai. [Q647, Q658]
- Group = vertical column; elements ki outermost shell electrons same hoti hain (valence electrons). [Q658, Q660]
- Lanthanides = 6th period ke baad; Actinides = 7th period ke baad; donon ko separately neeche rakha jaata hai taaki table compact rahe. [Q807, Q809]
- Inner transition elements = f-block; Lanthanides aur Actinides. [Q809, Q811]
- Metals left side aur center mein; non-metals right side mein (top-right corner mein); metalloids staircase/zig-zag line par jo metals ko non-metals se separate karti hai. Hydrogen ek exception hai — left side mein hone ke baawajood non-metal hai (sabse left-side ka akela non-metal). [Q811, Q813, Q784, Q864, Q936, Q974, Q980]
- Group 18 (Noble gases) ki valency zero hoti hai — inka valence shell already complete hota hai (octet, Helium ka duplet), isliye electrons na gain karte hain na lose. [Q665, Q790]

3. Hydrogen & Its Position

- Hydrogen ka atomic number 1; configuration $1s^1$. [Q829, Q831]
- Group 1 ke saath similarities: 1 valence electron; +1 oxidation state; electropositive. [Q831, Q833]
- Group 17 ke saath similarities: 1 electron short of noble gas; -1 oxidation state (hydride); diatomic molecule (H_2). [Q833, Q835]
- Isliye hydrogen ko alag se rakha jaata hai; sometimes center mein, sometimes donon groups ke upar. Electronic configuration alkali metals se milti-julti hai lekin hydrogen khud alkali metal NAHI hai; Mendeleev's periodic table mein hydrogen ki koi fixed position nahi thi (uncertain position, ye ek limitation thi). Halogens ki tarah hydrogen bhi diatomic molecule (H_2) ke roop mein exist karta hai. [Q835, Q837, Q860, Q725]
- Hydrogen metal nahi hai; non-metal hai; Left side ka akela non-metal hai. [Q837]
- ⚠ Trap: Hydrogen sometimes Group 1 ke upar rakha jaata hai lekin ye metal nahi hai. [Q751]

4. Periodic Trends Across Periods & Groups

- Atomic number increases, nuclear charge increases. [Q647, Q659]
- Valence electrons increase (1 se 8 tak). [Q647]
- Atomic radii decrease (kyunki nuclear charge ka pull badhta hai, electrons same shell mein hain). Order example: $B < Be < Li$ (period 2 mein left se right ghatta hai). [Q647, Q659, Q667, Q671, Q741, Q780]
- Metallic character decreases, non-metallic character increases — Group 16 mein Oxygen sabse zyada non-metallic hai. [Q659, Q661, Q649]
- Electropositivity decreases, electronegativity increases (Na se Cl tak non-metallic character badhta hai). [Q659, Q976]
- Ionization energy increases (generally, with exceptions). [Q659, Q661]
- Chemical reactivity: Period ke start mein metals zyada reactive (Group 1); middle mein kam; end mein non-metals

reactive (Group 17); noble gases (Group 18) least reactive. [Q647, Q661]

- Atomic number increases, new shells add hoti hain. [Q660, Q662]
- Valence electrons same rehte hain — isliye same group ke elements similar chemical properties rakhte hain aur same valency hoti hai. [Q660, Q655, Q914, Q958]
- Atomic radii increase (kyunki new shells add hote hain, shielding effect badhta hai). Example: Cs > Rb > K > Na > Li (Group 1 mein); I (largest) > Br > Cl > F (Group 17 mein); Rb sabse metallic Period 5 mein (sabse left position). [Q660, Q662, Q653, Q711, Q839, Q988, Q904]
- Metallic character increases (generally; for non-metals, reactivity with metals increases). Example: Ca > Sc > Ga > Ge (Period 4 mein left se right metallic character ghatta hai). [Q662, Q664, Q693, Q926, Q731, Q892]
- Electropositivity increases (metals ke liye). [Q662]
- Ionization energy decreases (kyunki outermost electrons nucleus se door hote hain). [Q662, Q664]
- Down a group, chemical reactivity: Metals ke liye reactivity increases (Fr sabse reactive metal); Non-metals ke liye reactivity decreases (F sabse reactive non-metal). [Q664, Q666]
- Atomic radius = nucleus se outermost electron shell ki distance. [Q668, Q670, Q672]
- Li sabse bada radius 2nd period mein; F sabse chhota. [Q647]
- Cs (Caesium) sabse bada atomic radius known element mein (excluding francium). [Q674]
- Cation (positive ion) ka radius parent atom se chhota hota hai — electron loss se same number of protons kam electrons hain; zyada pull per electron. [Q676, Q678]
- Anion (negative ion) ka radius parent atom se bada hota hai — electron gain se electron-electron repulsion badhta hai; cloud expand hota hai. [Q678, Q680]
- Isoelectronic species mein (same electrons): jyada nuclear charge → chhota radius. Example: $N^{3-} > O^{2-} > F^- > Ne > Na^+ > Mg^{2+} > Al^{3+}$ (sabke paas 10 electrons). [Q680, Q682]
- ⚠ Trap: Cation radius < Atomic radius < Anion radius — for same element. [Q676, Q678]
- Ionization energy = gaseous atom se outermost electron remove karne mein lagne wali energy; unit KJ/mol. [Q684, Q686, Q688, Q952]
- Across a period: IE increases (generally) — nuclear charge badhta hai; atomic radius ghatta hai; electrons tightly held hote hain. [Q686, Q688]
- Down a group: IE decreases — atomic radius badhta hai; outermost electrons door hain; shielding effect; isliye zyada atomic radius wale element (jaise Fe) electrons easily lose karte hain. [Q688, Q690, Q928]
- Exceptions: Group 2 > Group 13 (p-orbital mein electron easily remove hota hai); Group 15 > Group 16 (half-filled stability Group 15 mein). [Q690, Q692]
- Helium (He) ka sabse zyada IE; Francium (Fr) ka sabse kam. [Q692, Q694]
- 2nd IE > 1st IE hamesha (kyunki positive ion se electron remove karna aur mushkil). [Q694]
- Electron affinity = gaseous atom mein electron add karne par release hone wali energy. [Q696, Q698, Q700]
- Across a period: EA generally increases (non-metals ke liye) — nuclear charge badhta hai; added electron tightly held hota hai. [Q698, Q700]
- Down a group: EA generally decreases — atomic radius badhta hai; added electron door se attract hota hai. [Q700, Q702]
- Exceptions: Group 2 aur Group 15 ka EA kam hota hai (full/half-filled stability); Noble gases ka EA ≈ 0 ya positive (unfavorable). [Q702, Q704]
- Chlorine (Cl) ka sabse zyada EA (not fluorine, kyunki F mein electron-electron repulsion zyada); Order: Cl > F > Br > I. Cl ka electron affinity 3rd period mein sabse zyada hota hai. [Q704, Q706, Q848, Q942, Q992]
- More negative EA value = more energy released = more favorable. [Q706]
- Electronegativity = atom ka tendency shared electron pair ko apni taraf khinchne ki; Pauling scale se measure hoti hai. [Q708, Q710, Q712]
- Across a period: Electronegativity increases — nuclear charge badhta hai; same size mein zyada pull. [Q710, Q712]
- Down a group: Electronegativity decreases — atomic size badhta hai; shared electrons door hote hain (Order: Cl > Br > I). [Q712, Q714, Q804]

- Fluorine (F) sabse zyada electronegative (Pauling scale: 4.0); outermost config ns^2np^5 . Among Cl, S, Al, Mg diye options mein Cl sabse zyada electronegative hota hai. [Q714, Q716, Q681, Q768, Q934]
- Francium (Fr) sabse kam electronegative. [Q716]
- Electronegativity difference se bond type predict hota hai: Difference $> 1.7 \rightarrow$ Ionic bond; Difference $< 1.7 \rightarrow$ Covalent bond. [Q716, Q718]
- Across a period: Metallic character decreases, non-metallic character increases. [Q718, Q720, Q722]
- Down a group: Metallic character increases, non-metallic character decreases. [Q722, Q724]
- Most metallic: Francium (Fr) (theoretical); Caesium (Cs) (practically observed). [Q724]
- Most non-metallic: Fluorine (F). [Q724]
- Metalloids (B, Si, Ge, As, Sb, Te) staircase line ke paar hote hain; properties intermediate. [Q726]
- Across a period: Valency pehle badhti phir ghatti — Group 1-4: valency $1 \rightarrow 4$ (electrons donate); Group 4-8: valency $4 \rightarrow 1$ (electrons accept, 8-n rule). Electropositive elements (metals) generally 1, 2, ya 3 valency rakhte hain. [Q728, Q730, Q788]
- Down a group: Valency same rehti hai; outermost electrons same hote hain. [Q730, Q732]
- Transition elements ke multiple oxidation states hote hain (variable valency). Example: Fe^{2+} aur Fe^{3+} ; Cu^+ aur Cu^{2+} . [Q732, Q734]
- Metallic character across period: decreases; down group: increases (for metals). [Q813, Q815, Q817]
- Non-metallic character across period: increases; down group: decreases. [Q815, Q817]
- Chemical reactivity of metals across period: first decreases (Group 1 $\rightarrow$ Group 14), phir increases (Group 15 $\rightarrow$ Group 17 as non-metals); down group: increases (for alkali metals). [Q817, Q819]
- Chemical reactivity of non-metals across period: increases; down group: decreases (halogens mein F sabse reactive). [Q819, Q821]
- Across a period: Oxides ka nature basic $\rightarrow$ amphoteric $\rightarrow$ acidic hota hai. [Q821, Q823, Q825]
- Examples (Period 3): Na_2O = strongly basic; MgO = basic; Al_2O_3 = amphoteric; SiO_2 = weakly acidic; P_4O_{10} = acidic; SO_3 = strongly acidic; Cl_2O_7 = strongly acidic. [Q823, Q825]
- Down a group: Same oxidation state mein oxides ka basic character badhta hai. Example: N_2O_5 (acidic) $< P_4O_{10}$ (acidic) $< As_2O_5$ (weakly acidic) $< Sb_2O_5$ (amphoteric) $< Bi_2O_5$ (basic). Alkaline oxide formation tendency down the group increases. [Q825, Q827, Q679]
- Higher oxidation state ke oxides zyada acidic hote hain. Example: SO_2 (weakly acidic) $< SO_3$ (strongly acidic). [Q827, Q829]

5. s-Block Elements (Groups 1 & 2)

- s-Block = Groups 1 aur 2; valence electrons ns^1 ya ns^2 mein hote hain. [Q751, Q753, Q755, Q830]
- Sabse zyada reactive metals; water ke saath vigorously react karte hain H_2 gas dete hain. Potassium alkali metal family ka member hai. [Q753, Q755, Q757, Q774]
- +1 oxidation state. [Q753]
- Soft metals (knife se cut kar sakte hain); low melting/boiling points. [Q883, Q885, Q887]
- Reactivity down the group badhti hai (ionization energy ghatti hai); Sodium apne se niche ke Ca, Zn, Mg se zyada reactive hai (reactivity series mein upar). [Q753, Q755, Q757, Q948]
- Melting point down the group decreases (atomic size badhne se metallic bond kamzor hoti hai). [Q885, Q887]
- Density generally increases down the group (exception: $Na > K$). [Q887, Q889]
- Water ke saath vigorous reaction; H_2 gas aur metal hydroxide. $2Na + 2H_2O \rightarrow 2NaOH + H_2 \uparrow$ (exothermic). [Q889, Q891]
- Potassium (K) water par float karta hai aur burn karta hai (lilac flame); sodium bhi float karta hai. [Q891, Q893]
- Cesium (Cs) aur Francium (Fr) explosive reaction water ke saath. [Q893, Q895]

- Oxides: Li_2O (normal oxide), Na_2O (normal oxide), K/Rb/Cs superoxide bhi bana sakte hain (KO_2). [Q895, Q897]
- Flame test colours: Li (crimson red), Na (golden yellow), K (lilac/violet), Rb (red-violet), Cs (blue). [Q753, Q755, Q757]
- Less reactive than alkali metals; water ke saath slowly react. Group 2 ko "alkaline earth metals" kehte hain. [Q755, Q757, Q759, Q930]
- +2 oxidation state. [Q755]
- Harder than alkali metals; higher melting points. [Q897, Q899, Q901]
- Down the group: reactivity increases; melting point decreases; density increases. [Q899, Q901]
- Beryllium (Be) anomalous: covalent compounds; other ionic; Be aur atomic number 20 wala element (Calcium) similar chemical properties dikhate hain (same group 2). [Q899, Q816]
- Flame test colours: Ca (brick red), Sr (crimson), Ba (apple green). [Q755, Q757, Q759]
- Magnesium (Mg): lightweight; alloys (magnalium); chlorophyll ka center; flashbulbs. [Q901, Q903]
- Calcium (Ca): bones/teeth; lime (CaO); cement; water hardness. [Q903, Q905]
- Barium (Ba): BaSO_4 X-ray contrast agent (barium meal); BaCO_3 rat poison. [Q905, Q907]
- Radium (Ra): radioactive; historical cancer treatment; luminescent paints. [Q907, Q909]

6. p-Block Elements (Groups 13-18)

- p-Block = Groups 13 se 18; valence electrons $ns^2 np^{1-6}$ mein hote hain. [Q759, Q761, Q763]
- +3 oxidation state (B exceptions bhi bana sakta hai). [Q761, Q763]
- Boron = metalloid (first metalloid in periodic table); baaki metals. Aluminium (13) aur Gallium (31) same group (13) ke elements hain. [Q761, Q763, Q842, Q906]
- +4 aur +2 oxidation states; down the group +2 stable hota hai (inert pair effect). [Q763, Q765]
- Carbon aur Silicon non-metals; Germanium metalloid; Tin aur Lead metals — is group mein sirf Carbon non-metal hai. [Q763, Q765, Q701, Q719]
- -3, +3, +5 oxidation states; down the group metallic character badhta hai. [Q765, Q767]
- Nitrogen aur Phosphorus non-metals; Antimony aur Bismuth metals. Nitrogen family ka 3rd element (N ke baad) Arsenic hai (atomic number 33). [Q765, Q767, Q846]
- -2 oxidation state (generally); +4 aur +6 bhi. [Q767, Q769]
- Oxygen aur Sulphur non-metals; Polonium metal. [Q767, Q769]
- -1 oxidation state; most reactive non-metals; Group number = 17. Carbon family (Group 14) ke saath compare karte hain group numbers. [Q769, Q771, Q773, Q705, Q794]
- Reactivity down the group ghatti hai (F sabse reactive). [Q769, Q771, Q773]
- Down the group: atomic radius increases; electronegativity decreases; electron affinity generally decreases (Cl max). [Q867, Q869, Q871]
- Physical state: F_2 (pale yellow gas) $\rightarrow$ Cl_2 (greenish-yellow gas) $\rightarrow$ Br_2 (red-brown liquid) $\rightarrow$ I_2 (violet-black/purple solid — sabse dark colour halogen). [Q769, Q771, Q773, Q871, Q873, Q924]
- Reactivity: $\text{F}_2 \gg \text{Cl}_2 > \text{Br}_2 > \text{I}_2$; F_2 sabse zyada oxidizing. [Q873, Q875]
- Displacement: Upward halogen downward halogen ke salt se displace kar sakta hai. $\text{Cl}_2 + 2\text{NaBr} \rightarrow 2\text{NaCl} + \text{Br}_2$. [Q875, Q877]
- Interhalogen compounds: ClF , BrF_3 , ICl , etc. [Q877, Q879]
- HF weak acid (hydrogen bonding); HCl, HBr, HI strong acids. [Q879, Q881]
- Bleaching powder (CaOCl_2) chlorine se banta hai; disinfectant. [Q881, Q883]
- Halogen group ki oxygen ke saath maximum valency 7 hoti hai (Cl_2O_7 example). Atomic number 17 (Cl) atomic number 35 (Br) ke saath resemble karta hai (same group 17). [Q792, Q922, Q902]
- Fire extinguisher mein Bromine, dental health mein Fluorine, refrigerants mein halogens, thyroid hormones mein

Iodine use hota hai. [Q984]

- Valence shell complete ($ns^2 np^6$, He mein $1s^2$); chemically inert (generally); "VIII A" bhi kaha jaata hai purane naming mein. He, Ne, Ar teenon ki outermost shell filled hoti hai. [Q773, Q775, Q851, Q853, Q855, Q822, Q894, Q900, Q916]
- Down the group: atomic radius increases; boiling point increases (weak van der Waals forces; larger atoms = more polarizable). [Q853, Q855]
- Helium (He): sabse light; non-flammable; balloons, blimps, diving mixtures; liquid He cryogenics. [Q855, Q857]
- Neon (Ne): red discharge tubes; advertising signs; high-voltage indicators. [Q857, Q859]
- Argon (Ar): sabse abundant noble gas atmosphere mein; light bulbs mein fill karte hain (filament ko oxidize hone se bachane); welding shield gas; Ar ka electron configuration (2,8,8) mein M-shell (outer cell) mein 3 orbitals mein 8 electrons hote hain. [Q859, Q861, Q834]
- Krypton (Kr): high-performance light bulbs; lasers; Period 4 ka inert gas (atomic number 36). [Q861, Q863, Q841]
- Xenon (Xe): headlights (xenon lamps); anesthesia (historical); spacecraft ion propulsion. [Q863, Q865]
- Radon (Rn): radioactive; uranium decay se; cancer treatment (radiotherapy) mein use historically; lung cancer risk bhi. [Q865, Q867]
- ⚠ Trap: Noble gases "inert" historically kehte the lekin ab kuch compounds bana sakte hain (XeF_2 , XeF_4 , XeF_6 — Bartlett, 1962). [Q867]

7. d-Block & f-Block Elements

- d-Block = Groups 3 se 12; transition metals; valence electrons $(n-1)d^{1-10} ns^{0-2}$ mein; two outermost shells incomplete hoti hain — isliye variable oxidation states (variable valency) exhibit karte hain. [Q775, Q777, Q779, Q852, Q932, Q966]
- Characteristics: Variable oxidation states (multiple valency); Coloured compounds/ions (due to d-d transitions); Catalytic properties; Formation of complex compounds; Magnetic properties. [Q777, Q779, Q781]
- Examples: Fe, Cu, Zn, Mn, Ni, Co, Cr, Ti, V. [Q779, Q781]
- f-Block = Lanthanides (4f) aur Actinides (5f); inner transition metals; inko periods 6 aur 7 mein classify karte hain; teeno outermost shells incomplete hoti hain (rare earth elements characteristic). [Q781, Q783, Q854, Q910]
- Lanthanides (Ce se Lu): similar properties; +3 oxidation state common. Cerium (Ce, atomic number 58) Lanthanide series ka element hai, f-block mein aata hai. [Q783, Q818, Q820]
- Actinides (Th se Lr): radioactive; uranium aur plutonium nuclear fuel mein. Californium (Cf) bhi Actinide series ka member hai; sab radioactive elements is series mein aate hain. [Q783, Q785, Q826, Q876]

8. Representative vs Transition Elements

- Representative elements = s-block aur p-block (Groups 1, 2, 13-18); valence electrons outermost shell mein. [Q785, Q787]
- Transition elements = d-block (Groups 3-12); valence electrons penultimate shell ke d-orbitals mein bhi hote hain. [Q787, Q789]
- Representative elements ke paas generally fixed valency hoti hai; transition elements mein variable valency. [Q789, Q791]
- Representative elements ke compounds generally colourless hote hain (exceptions hain); transition elements ke compounds coloured hote hain. [Q791, Q793]

9. Diagonal Relationship

- Diagonal relationship = Period mein ek element aur uske next period ke next group ke element mein similar

properties. [Q793, Q795, Q797]

- Examples:
 - Li (Group 1, Period 2) aur Mg (Group 2, Period 3): donon nitrates heat par decompose hote hain; hydroxides less soluble.
 - Be (Group 2, Period 2) aur Al (Group 13, Period 3): donon amphoteric oxides banate hain; chlorides covalent.
 - B (Group 13, Period 2) aur Si (Group 14, Period 3): donon metalloids; acids se react karte hain. [Q795, Q797, Q799]
- Reason: Similar ionic size/charge ratio; similar polarizing power. [Q799]

10. Anomalous Properties of First Elements

- Har group ka first element (Period 2) generally apne group ke baaki elements se alag properties rakhta hai. [Q799, Q801, Q803]
- Reasons: Small atomic size; High electronegativity; No d-orbitals available for bonding (except 2nd period elements); High ionization energy. [Q801, Q803]
- Examples: Li vs other alkali metals; Be vs other alkaline earth metals; B vs other group 13 elements; C vs other group 14 elements; N vs other group 15 elements; O vs other group 16 elements; F vs other halogens. [Q803, Q805]

11. Metalloids & Their Uses

- Metalloids = properties intermediate between metals and non-metals; semiconductors. [Q909, Q911, Q913]
- Common metalloids: Boron (B), Silicon (Si), Germanium (Ge), Arsenic (As), Antimony (Sb), Tellurium (Te) — total 7 metalloids modern periodic table mein (staircase line ke around). Period 4 mein Ge aur As metalloids hain; Period 5 mein Sb aur Te. [Q911, Q913, Q872]
- Silicon (Si): semiconductors; solar cells; computer chips; silica (SiO_2) = glass, quartz; silicon hydrogen ke saath chain compounds (silanes) bhi banata hai jinmein 7-8 atoms ki chain hoti hai (weak Si-H bonds ki wajah se reactive). [Q913, Q915, Q960]
- Germanium (Ge): early transistors; fiber optics; infrared optics; semiconductor bhi hai. [Q915, Q917, Q850]
- Boron (B): borosilicate glass (Pyrex); borax ($\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$); neutron capture (cancer therapy). [Q917, Q919]
- Arsenic (As): pesticides; semiconductors (GaAs). [Q919, Q921]
- Antimony (Sb): alloys (type metal, Babbitt metal); flame retardants. [Q921, Q923]
- Tellurium (Te): alloys (steel/machinery); solar panels (CdTe); thermoelectric devices. [Q923, Q925]

12. Important Chemical Reactions & Compounds

- Aqua regia = conc. HNO_3 + conc. HCl (1:3); gold aur platinum dissolve kar sakta hai. [Q925, Q927]
- Chlor-alkali process: $2\text{NaCl} + 2\text{H}_2\text{O} \rightarrow 2\text{NaOH} + \text{Cl}_2 + \text{H}_2$ (electrolysis). [Q927, Q929]
- Contact process: Sulphuric acid manufacturing; $\text{SO}_2 \rightarrow \text{SO}_3$ (V_2O_5 catalyst). [Q929, Q931]
- Haber process: $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$ (Fe catalyst, 400–500°C, 200 atm); ammonia fertilizers ke liye. [Q931, Q933]
- Ostwald process: $\text{NH}_3 \rightarrow \text{NO} \rightarrow \text{NO}_2 \rightarrow \text{HNO}_3$ (Pt catalyst); nitric acid manufacturing. [Q933, Q935]
- Bayer process: Bauxite se alumina purify karna (NaOH se); aluminium extraction ke liye. [Q935, Q937]
- Hall-Héroult process: Alumina ka electrolysis cryolite mein; aluminium metal. [Q937, Q939]

13. Comparative Chemistry: Important Pairs

- Mg vs Ca: $\text{Mg}(\text{OH})_2$ weak base; $\text{Ca}(\text{OH})_2$ strong base; Mg forms basic carbonate; Ca nahi. [Q951, Q953]
- Na vs K: Na^+ aur K^+ biological systems mein important (Na-K pump); K more reactive than Na in some contexts. [Q953, Q955]
- Li vs Na: Li unique (diagonal relationship with Mg); Li compounds more covalent. [Q955, Q957]
- C vs Si: CO_2 gas, molecular; SiO_2 solid, network covalent; Si larger, forms Si-Si bonds less readily. [Q957, Q959]
- N vs P: N_2 triple bond, inert; P_4 tetrahedral, reactive; N forms $\pi\pi$ - $\pi\pi$ bonds, P less readily. [Q959, Q961]
- O vs S: O_2 paramagnetic; O_3 diamagnetic; S_8 crown structure; H_2O hydrogen bonding, H_2S nahi. [Q961, Q963]

14. Everyday Applications & Industrial Uses

- Silicon chips = computers, smartphones. [Q963, Q965]
- Aluminium = aircraft, utensils, foils (lightweight, corrosion resistant). [Q965, Q967]
- Iron/Steel = construction, vehicles, machinery. [Q967, Q969]
- Copper = electrical wires (excellent conductor). [Q969, Q971]
- Titanium = aerospace, medical implants (biocompatible, strong). [Q971, Q973]
- Chromium = stainless steel, electroplating (shiny finish). [Q973, Q975]
- Nickel = batteries (Ni-Cd, Ni-MH, Li-ion mein bhi), alloys. [Q975, Q977]
- Silver = jewelry, photography (historical), antibacterial. [Q977, Q979]
- Gold = jewelry, electronics (non-reactive, conductive); reactivity series mein sabse kam reactive metals mein se ek (bottom mein). [Q979, Q981, Q742]
- Mercury = thermometers (historical), dental amalgams, electrical switches. [Q981, Q983]
- Carbon apne tetravalency aur catenation ki wajah se sabse zyada compounds banata hai (organic + inorganic dono). [Q918]
- Bromine akela non-metal hai jo liquid state mein exist karta hai room temperature par (Mercury metal hai jo liquid hota hai). [Q946]
- Eugenol = cloves (laung) mein paya jaane wala essential oil; perfume aur spice dono mein use hota hai. [Q896]
- Benzene (C_6H_6) ek unsaturated hydrocarbon hai. [Q878]
- Osmium (Os) sabse zyada density wala element hai. [Q764]

15. Radioactive & Synthetic Elements

- Technetium (Tc, Z=43): sabse halka element jo naturally stable nahi hai; first artificial element. [Q983, Q985]
- Promethium (Pm, Z=61): rare earth; radioactive; no stable isotopes. [Q985, Q987]
- Actinides: sab radioactive; Thorium aur Uranium naturally abundant; baaki synthetic/transuranic. Titanium radioactive NAHI hai (stable element hai) — U, Th, Ac radioactive hain. [Q987, Q989, Q968]
- Transuranic elements: atomic number > 92; synthetic; particle accelerators/se nuclear reactors mein banaye jaate hain. Neptunium (Np, 93), Plutonium (Pu, 94) — nuclear weapons/fuel. Einsteinium (Es, 99), Fermium (Fm, 100) — named after scientists. [Q989, Q991]
- Oganesson (Og, 118): sabse heavy confirmed element; synthetic; named after Yuri Oganessian. [Q991, Q993]

16. Memory Aids & Positions

- Period 2 elements: Li, Be, B, C, N, O, F, Ne — "Lina Be Boli Cini Nahi Oxygen Flouride Ne". [Q939, Q941]
- Group 1: Li, Na, K, Rb, Cs, Fr — "Lina Na Kali Rubi Cesi France". [Q941, Q943]
- Group 17: F, Cl, Br, I, At — "Fiza Chlorine Bharati Iodine Aati". [Q943, Q945]
- First 30 elements ke atomic numbers aur symbols yaad rakhna zaroori hai exam ke liye — jaise Lead (Pb), Zinc (Zn), Iron (Fe), Magnesium (Mg), Sodium (Na), Aluminium (Al), Sulphur (S). [Q945, Q947, Q756]
- Transition metals (d-block): Sc(21) se Zn(30) tak 4th period; Y(39) se Cd(48) tak 5th period. [Q947, Q949]
- Inner transition: Lanthanides (57–71), Actinides (89–103). [Q949, Q951]

17. ⚠ Traps / Key Differences Summary

- ⚠ Trap: Atomic radius period mein ghatta hai lekin group mein badhta hai — opposite trends hain. [Q647, Q670, Q672]
- ⚠ Trap: Ionization energy ke exceptions hain: Group 2 > Group 13, Group 15 > Group 16 — half-filled/full stability. [Q690, Q692]
- ⚠ Trap: Chlorine ka EA sabse zyada hai, Fluorine ka nahi — F mein electron-electron repulsion kam favorable banata hai. [Q704, Q706]
- ⚠ Trap: Electronegativity difference > 1.7 → ionic; < 1.7 → covalent — ye threshold yaad rakhna. [Q716, Q718]
- ⚠ Trap: Metallic character ke liye Fr theoretically sabse metallic hai lekin practically Cs observe kiya jaata hai. [Q724]
- ⚠ Trap: Valency period mein 1→4→1 pattern follow karti hai (electrons donate → share → accept). [Q728, Q730]
- ⚠ Trap: Dobereiner triads, Newlands octaves, Mendeleev periodic table — teenon ke limitations alag hain; exam mein match karte hain; teenon hi atomic mass ke basis par arrange karte the (atomic number par nahi). [Q648, Q650, Q652, Q858]
- ⚠ Trap: Alkali metals ka reactivity down the group badhti hai (ionization energy ghatti); Halogens ka reactivity ghatti hai (electron affinity kam hoti hai). [Q753, Q771]
- ⚠ Trap: Flame test colours yaad rakhna: Na = yellow, K = lilac, Ca = brick red, Ba = green — exam mein direct puchha jaata hai. [Q757, Q759]
- ⚠ Trap: Transition metals ke variable oxidation states hote hain; representative elements ke generally fixed. [Q777, Q789]
- ⚠ Trap: Diagonal relationship ke examples: Li-Mg, Be-Al, B-Si — properties similar hain lekin groups alag hain. [Q795, Q797]
- ⚠ Trap: First element of each group (Period 2) generally anomalous properties rakhta hai apne group se — small size aur high electronegativity ki wajah se. [Q801, Q803]
- ⚠ Trap: Oxides ka nature period mein basic → amphoteric → acidic hota hai; ye trend important hai. [Q823, Q825]
- ⚠ Trap: Higher oxidation state ke oxides zyada acidic hote hain; same element ke liye compare karne par. [Q827, Q829]
- ⚠ Trap: Hydrogen ko Group 1 aur Group 17 donon ke saath compare karte hain; isliye unique position hai. [Q831, Q833]
- ⚠ Trap: Noble gases ab "inert" nahi kehte — Xe ke fluorides (XeF₂, XeF₄, XeF₆) hain; Neil Bartlett (1962) ne pehla noble gas compound banaya. [Q867]
- ⚠ Trap: Halogens ka physical state down the group change hota hai: gas → liquid → solid; colour bhi dark hoti hai. [Q871, Q873]
- ⚠ Trap: HF weak acid hai hydrogen bonding ki wajah se; HCl, HBr, HI strong acids hain. [Q879, Q881]
- ⚠ Trap: Alkali metals ka melting point down the group ghatta hai; density generally badhti hai (exception Na > K). [Q885, Q887]
- ⚠ Trap: Aqua regia = HNO₃:HCl (1:3) — gold aur platinum dissolve karta hai; sirf HCl ya HNO₃ se alag nahi hota. [Q925, Q927]

- ⚠ Trap: Contact process mein V_2O_5 catalyst use hota hai $SO_2 \rightarrow SO_3$ conversion ke liye. [Q929, Q931]
- ⚠ Trap: Haber process mein Fe catalyst, 400–500°C, 200 atm — conditions yaad rakhna. [Q931, Q933]
- ⚠ Trap: Silicon (Si) semiconductor hai; Silica (SiO_2) glass/quartz hai — donon alag. [Q913, Q915]
- ⚠ Trap: Barium meal mein $BaSO_4$ use hota hai X-ray ke liye; Ba^{2+} toxic hai lekin $BaSO_4$ insoluble hone ki wajah se safe. [Q905]
- ⚠ Trap: Technetium (Z=43) aur Promethium (Z=61) — donon naturally stable isotopes nahi rakhte. [Q983, Q985]
- ⚠ Trap: Transuranic elements (Z > 92) sab synthetic hain; naturally nahi milte. [Q989, Q991]

18. Electronic Configuration → Element Identification (Practice)

- Config 2, 6 (total 8 electrons) = Oxygen (atomic number 8). [Q880, Q956]
- Config 2, 8, 2 = Magnesium (atomic number 12); Period 3 mein; valency +2 (2 electrons lose karke noble gas config paata hai). [Q798, Q824]
- Config 2, 8, 3 = Aluminium (atomic number 13); 3 valence electrons. [Q750]
- Config 2, 8, 4 = Silicon (atomic number 14). [Q715]
- Config 2, 8, 5 = Phosphorus (atomic number 15). [Q938]
- Config 2, 8, 6 = Sulphur (atomic number 16); Group 16, Period 3. [Q866]
- Config 2, 8, 7 = Chlorine (atomic number 17). [Q754]
- Config 2, 8, 8 = Argon (atomic number 18); Group 18 (noble gas). [Q729, Q752]
- Carbon (atomic number 6) ki L-shell mein 4 electrons hote hain (K-shell mein 2). [Q677]
- Rhodium (Rh, Z=45) ka full electronic configuration example bhi exam mein puchha jaata hai — silver-white, corrosion-resistant metal (1803 mein William Wollaston ne discover kiya). [Q762]

19. Shell Capacity Rule ($2n^2$) & Basic Structure Facts

- Formula: shell mein max electrons = $2n^2$ (n = shell number). K-shell (n=1) = 2, L-shell (n=2) = 8, M-shell (n=3) = 18, N-shell (n=4) = 32. [Q828, Q836, Q845]
- Valence electrons = atom ki outermost shell mein present electrons. [Q760]
- Period 3 ke elements (Na se Ar) ke paas 3 shells hote hain: K, L, M. [Q802]
- Atomic number = nucleus mein protons ki sankhya; ye element ki periodic table mein position determine karta hai (Rutherford ne 1917 mein proton discover kiya tha). [Q766]
- Isotopes: same atomic number (same protons) lekin different mass number; different physical properties (mass alag hone se) lekin similar chemical properties (same electron config hone se). Protium (H isotope) mein 0 neutrons hote hain — sirf 1 proton aur 1 electron. [Q940, Q758]

20. Period-wise & Group-wise Element Counts

- Period 1 = sirf 2 elements (Hydrogen, Helium) — dono gases hain. Period 2 aur 3 = 8-8 elements each. Period 4 aur 5 = 18-18 elements each. Period 6 aur 7 = 32-32 elements each. [Q886, Q972, Q890, Q944, Q814]
- Total 18 groups aur 7 periods modern periodic table mein hote hain. [Q847]

21. Family Names, Block Classification & Rare Earth Facts

- Group 1 = Alkali metals (Li, Na, K, Rb, Cs, Fr) — highly reactive, 1 valence electron. [Q774, Q888, Q908, Q986]
- Group 2 = Alkaline earth metals (Be, Mg, Ca, Sr, Ba, Ra). [Q930]
- s-block = Group 1 aur 2 ke elements (last electron s-orbital mein enter karta hai). [Q830]
- d-block (Group 3-12) = Transition elements; variable oxidation states exhibit karte hain (jaise Cu, Fe, Ni). [Q852, Q932, Q966]
- f-block = Lanthanides aur Actinides = Rare earth elements; inke 3 outermost shells incomplete hote hain (4f/5d/6s ya 5f/6d/7s subshells). [Q818, Q820, Q826, Q854, Q910]
- Radioactive elements mainly Actinides series mein placed hain (Actinium se Lawrencium tak, Z=89-103). [Q876]

22. Atomic Number → Group/Period Matching Practice

- Atomic number 20 (Calcium) Be (atomic number 4) ke saath similar chemical properties dikhata hai — donon Group 2 mein hain. [Q816]
- Atomic number 13 (Aluminium) aur 31 (Gallium) same group (13) mein hain. [Q906]
- Atomic number 16 wala element 3rd period mein aata hai (config 2,8,6). [Q849]
- Atomic number 17 (Chlorine) atomic number 35 (Bromine) ke saath resemble karta hai — donon Group 17 (halogens) mein. [Q902]
- Atomic number 11 se 18 tak (Na se Ar) sab elements Period 3 mein aate hain. [Q740]
- Diye gaye atomic numbers mein sabse kam atomic number wala element period mein sabse zyada metallic hota hai (left-most position). [Q892, Q904]
- Atomic numbers 2, 3, 7, 10, 30 diye gaye — jo 3-10 range mein hain (3, 7, 10) wo second period mein same period share karte hain. [Q796]
- Diye gaye elements Sc, Ga, Be, As mein Sc (Scandium, Group 3) sabse zyada metallic character dikhata hai (period mein sabse left position). [Q739]
- B(5), Ne(10), K(19), O(8) ko increasing atomic number order mein arrange karne par: B, O, Ne, K. [Q733]

23. Miscellaneous Element Facts, Uses & Ionization Facts

- Gallium metal itna low melting point rakhta hai ki hatheli (palm) ki garmi se hi pighal jaata hai. [Q844]
- Germanium (Ge) ek semiconductor hai. [Q850]
- Silicon hydrogen ke saath silanes (chain compounds, 7-8 atoms tak) banata hai — inorganic analogue of alkanes, lekin weak Si-H bonds ki wajah se highly reactive. [Q960]
- Sodium apni high reactivity ki wajah se aasani se cation (Na^+) banata hai, electropositive metal hai. [Q948]
- Ionization energy ka SI unit KJ/mol hota hai. [Q952]

24. Additional Practice Facts (Position, Valency & Statements)

- P, Q, R, S elements atomic numbers 2, 6, 13, 18 — P (noble gas, valency 0) aur S (noble gas, valency 0) same valency rakhte hain. [Q673]
- Element ki colour uski periodic table position se directly governed NAHI hoti — baaki properties (radius, valency, valence electrons) position-dependent hoti hain. [Q683]
- Element X jo Chlorine ke saath XCl banata hai aur periodic table ke left side mein hai — X = Hydrogen (1 valence electron, non-metal, gas); HCl banata hai. [Q685]
- Period 2, Group 2 ka element (Beryllium) ke 2 valence electrons hote hain. [Q687, Q703]
- Match karna: smallest atom = Lithium; metalloid = Silicon; most metallic natural element = Cesium; most

electronegative = Fluorine; least electronegative = Iodine (halogens mein). [Q691, Q721]

- Atomic numbers 15, 16, 17, 18 = Phosphorus (P), Sulphur (S), Chlorine (Cl), Argon (Ar). [Q695, Q748]
- Eka-aluminium (Mendeleev ki predicted element) ka oxide = Ga_2O_3 (Gallium oxide). [Q697]
- Second period ke elements (Li se Ne) same number of shells (2) rakhte hain, lekin valence electrons alag-alag hote hain. [Q699]
- Tellurium (Z=52) periodic table mein Selenium (34, upar), Antimony (51, left), Iodine (53, right) aur Polonium (84, neeche) se surrounded hai. [Q707]
- ⚠ Statement check: Potassium reactivity series mein LEAST reactive nahi, balki sabse MOST reactive common metals mein se ek hai. [Q717]
- Fluorine sabse zyada tendency rakhta hai electron gain karne ki (sabse zyada electron affinity trend mein, lekin actual max EA Chlorine ka hai due to F ki repulsion). [Q745]
- Fluorine aur Chlorine dono ke 7 valence electrons hote hain (Group 17). [Q747]
- Zinc (Zn, atomic number 30) ki position: Group 12, Period 4 (d-block). [Q749]
- Period 1 mein sirf 2 elements hote hain (H aur He) — 1 element wala koi bhi period nahi hota. [Q770]
- Modern periodic table mein elements increasing nuclear charge (atomic number) ke order mein arranged hote hain. [Q772]
- Electron affinity down the group generally decrease hoti hai (top se bottom). [Q778]
- Group 18 ke elements (noble gases) ka outer shell completely filled hota hai. [Q786]
- Period 4, Group 2 ka element = Calcium (Ca). [Q810]
- Helium (He) ka maximum ionisation potential hota hai saare elements mein. [Q832]
- Alkali metal group ki characteristics: sab metals hain, water ke saath react karke H_2 gas dete hain, sabki valency 1 hoti hai. [Q838]
- Electronegativity = halogen atoms ki electrons attract karne ki property. [Q840]
- Mercurous ion (Hg^+) ki valency 1 hoti hai (Mercuric Hg^{2+} ki valency 2). [Q843]
- Same group ke elements same number of valence electrons rakhte hain, isliye same chemical properties dikhate hain. [Q856, Q950]
- Symbol H hydrogen element ko represent karta hai — lightest, colourless, odourless, diatomic gas. [Q868]
- Calcium ka atomic mass = 40 hota hai. [Q884]
- Statement check: Har ore ek mineral hota hai lekin har mineral ore nahi hota (ore mein economically extractable metal hona zaroori hai). [Q912, Q920]
- Be (4) aur B (5) ka pair increasing atomic number ka correct example hai (left to right). [Q982]